Stages 1–3: Financial News ⟶ Four Topic Readers ⟶ Narrative Attention & Shocks

1. Financial News Stream

1.26M articles, 2012–2024 (FinMultiTime, FNSPID)

ENERGY / COMMODITIES — MAR 2022
US Bans Russian Oil Imports

BIOTECH / HEALTH — NOV 2020
Vaccine Trial Shows 90% Efficacy

TECHNOLOGY — SEP 2023
Apple Unveils New iPhone Lineup

headline + lead; news of day d used from $d{+}1$

Articles

2. Four Topic Readers

fitted on text to 2019, applied forward

LDA — 60 topics
word counts, Dirichlet mixture

MiniLM — 60 clusters
sentence embeddings, k-means

FinBERT — 20 classes
headline topic classifier

NoLBERT — 60 / 20 clusters
1976–95 text, contrastively adapted

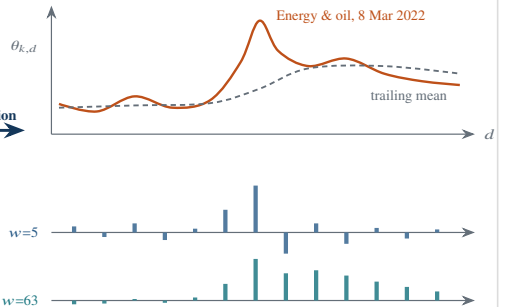
weights $\pi_{a,k} \geq 0$, $\sum_k \pi_{a,k} = 1$

Attention

$\theta_{k,d}$

3. Daily Attention & Shocks

$s_{k,d}^{(w)} = \theta_{k,d} - \frac{1}{w}\sum_{j=1}^{w} \theta_{k,d-j}$



63-day shocks: themes building over a quarter

Shocks $s_{k,d}$, firm news, tone + daily returns $r_{i,d}$ (CRSP, 1,000 largest stocks)

Stages 4–6: Stock Features ⟶ Return Forecasts ⟶ Portfolios

4. Stock Features

cross-sectional ranks in $[-1, 1]$, each period

	price	exposure	firm	tone
XOM	+0.4	+0.9	+0.1	+0.2
DAL	−0.3	−0.8	+0.1	−0.1
AAPL	+0.6	+0.1	+0.7	+0.5
JPM	+0.1	−0.2	−0.6	+0

$z_{i,t}$

exposure: EW covariance of $r_{i,d}$ with $s_{k,d}$ over 252 days; missing values taken from a random stock

5. Return Forecasts

target: next-period excess return, demeaned

OLS · Ridge · Lasso · IPCA · Sparse IPCA · XGBoost · Random forest · EBM

$\hat{y}_{i,t+1}$

+ ranked targets · ensemble of the eight

Combination across readers
equal-weighted ranks of 6 readers × 8 models

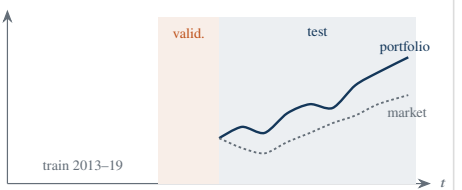
6. Portfolios & Evaluation

10 bp per dollar traded; test 2022–2024

decile long-short

buffered: hold within the fifth

cap-weighted legs · cap-weighted tilt



rank IC, net Sharpe, Ledoit–Wolf and SPA tests